

Advanced (Carolina Reaper)

Q1. Factor the expression $x^3 + 8$.

- (A) $(x + 2)(x^2 - 2x + 4)$
- (B) $(x - 2)(x^2 + 2x + 4)$
- (C) $(x + 2)(x^2 + 2x + 4)$
- (D) $(x - 2)(x^2 - 2x + 4)$
- (E) Cannot be factored

Q2. Factoring $27x^3 - 8y^3$ yields:

- (A) $(3x - 2y)(9x^2 + 6xy + 4y^2)$
- (B) $(3x + 2y)(9x^2 - 6xy + 4y^2)$
- (C) $(3x - 2y)(9x^2 - 6xy + 4y^2)$
- (D) $(3x + 2y)(9x^2 + 6xy + 4y^2)$
- (E) $(3x + 2y)(9x^2 - 6xy - 4y^2)$

Q3. What is the factored form of $64 - x^3$?

- (A) $(4 - x)(16 + 4x + x^2)$
- (B) $(4 + x)(16 - 4x + x^2)$
- (C) $(4 - x)(16 - 4x + x^2)$
- (D) $(4 + x)(16 + 4x + x^2)$
- (E) $(4 + x)(16 - 4x - x^2)$

Q4. A craft store sells $x^3 + 27$ boxes of beads. Factor this expression.

- (A) $(x + 3)(x^2 - 3x + 9)$
- (B) $(x - 3)(x^2 + 3x + 9)$
- (C) $(x + 3)(x^2 + 3x + 9)$
- (D) $(x - 3)(x^2 - 3x + 9)$
- (E) Cannot be factored

Q5. Factor $8x^3 - 1$.

- (A) $(2x - 1)(4x^2 + 2x + 1)$
- (B) $(2x + 1)(4x^2 - 2x + 1)$
- (C) $(2x - 1)(4x^2 - 2x + 1)$
- (D) $(2x + 1)(4x^2 + 2x + 1)$
- (E) $(2x - 1)(4x^2 + 2x - 1)$

Q6. If $a = 2x$ and $b = 3y$, factor $a^3 + b^3$.

- (A) $(2x + 3y)(4x^2 - 6xy + 9y^2)$
- (B) $(2x - 3y)(4x^2 + 6xy + 9y^2)$
- (C) $(2x + 3y)(4x^2 + 6xy + 9y^2)$
- (D) $(2x - 3y)(4x^2 - 6xy + 9y^2)$
- (E) $(2x - 3y)(4x^2 + 6xy - 9y^2)$

Q7. A city's population grows at a rate of $x^3 - 125$. Factor this expression.

- (A) $(x - 5)(x^2 + 5x + 25)$
- (B) $(x + 5)(x^2 - 5x + 25)$
- (C) $(x - 5)(x^2 - 5x + 25)$
- (D) $(x + 5)(x^2 + 5x + 25)$
- (E) $(x - 5)(x^2 + 5x - 25)$

Q8. Factor $1000x^3 + 8y^3$.

- (A) $(10x + 2y)(100x^2 - 20xy + 4y^2)$
- (B) $(10x - 2y)(100x^2 + 20xy + 4y^2)$
- (C) $(10x + 2y)(100x^2 + 20xy + 4y^2)$
- (D) $(10x - 2y)(100x^2 - 20xy + 4y^2)$
- (E) $(10x - 2y)(100x^2 + 20xy - 4y^2)$

Open Ended Questions (FRQ)

Q9. Factor the expression $x^3 + 64$ and explain how it relates to a real-life scenario, such as a hobby.

Q10. A time management app uses the expression $8x^3 - 1$ to calculate the time spent on tasks. Factor this expression and describe a possible daily life application.

Chef's Tips

- Use the factoring formulas $a^3 + b^3 = (a + b)(a^2 - ab + b^2)$ and $a^3 - b^3 = (a - b)(a^2 + ab + b^2)$
- Check for common factors before factoring
- Apply factoring to real-life scenarios to demonstrate problem-solving skills